Task 1: Geocoding Office Data

Initial Attempt

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Abstract

The following shows the first week progress on the geocoding office data task.

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1 This Week's Progress

We were able to run the US Census Geocoder on a subset of SF CoStar data (94% match rate), but encountered a scenario where places have "tie" entries (multiple coordinates found). Guidance on this is requested.

1.1 Tasks Completed

- Created Template
- Cleaned and Standardized SF adresses
- Use Census Geocoder to geocode addresses

1.2 Tasks In Progress

- Questions on handling errors

2 Data Work

2.1 Data Cleaning

All address-related variables are identified below Table 1.

Table 1: CoStar Variables Retained for Geocoding

Variable	Description
LeaseCompId	Unique lease ID used as the join key
PropertyId	Unique property identifier
PropertyName	Building name
Market	CoStar metro market
Submarket	CoStar submarket
Address.DeliveryAddress	Street address
Address.Locality	City
Address.County	County
Address.PostalCode	ZIP / postal code
Address.RegionName	State name (often missing; typically supplemented by Subdivision)
Address.Subdivision	State abbreviation
Address.Country	Country

2.2 Error Analysis

Table 2: Missing Address Summary

Missing Count	Missing (%)	Present Count
354	3.79	8,998

3 Issues and Questions

3.1 Geocoding Ambiguity: Directional Prefixes and Street-Type Suffixes

The Census Geocoder returned multiple equally-scored matches ("ties") for a subset of addresses. Two distinct sources of ambiguity were identified:

1. **Directional prefixes** : the same street number resolves to different segments depending on the cardinal direction prefix (e.g., 256 **S** EL CAMINO REAL, SAN MATEO, CA 94401 vs. 256 **N** EL CAMINO REAL, SAN MATEO, CA 94401).
2. **Street-type suffixes** : the same street number resolves to different street subtypes (e.g., 100 SKYLINE **PLZ**, DALY CITY, CA 94015 vs. 100 SKYLINE **DR**, DALY CITY, CA 94015).

Is there available directional information?

The Haversine distance between paired tie candidates ranges from 0 to approximately 6 km across the Bay Area sample. Figure 1 plots each tie pair connected by a line, with the distance labeled at the midpoint. An interactive version is available at `temp/tie_records_map.html`.

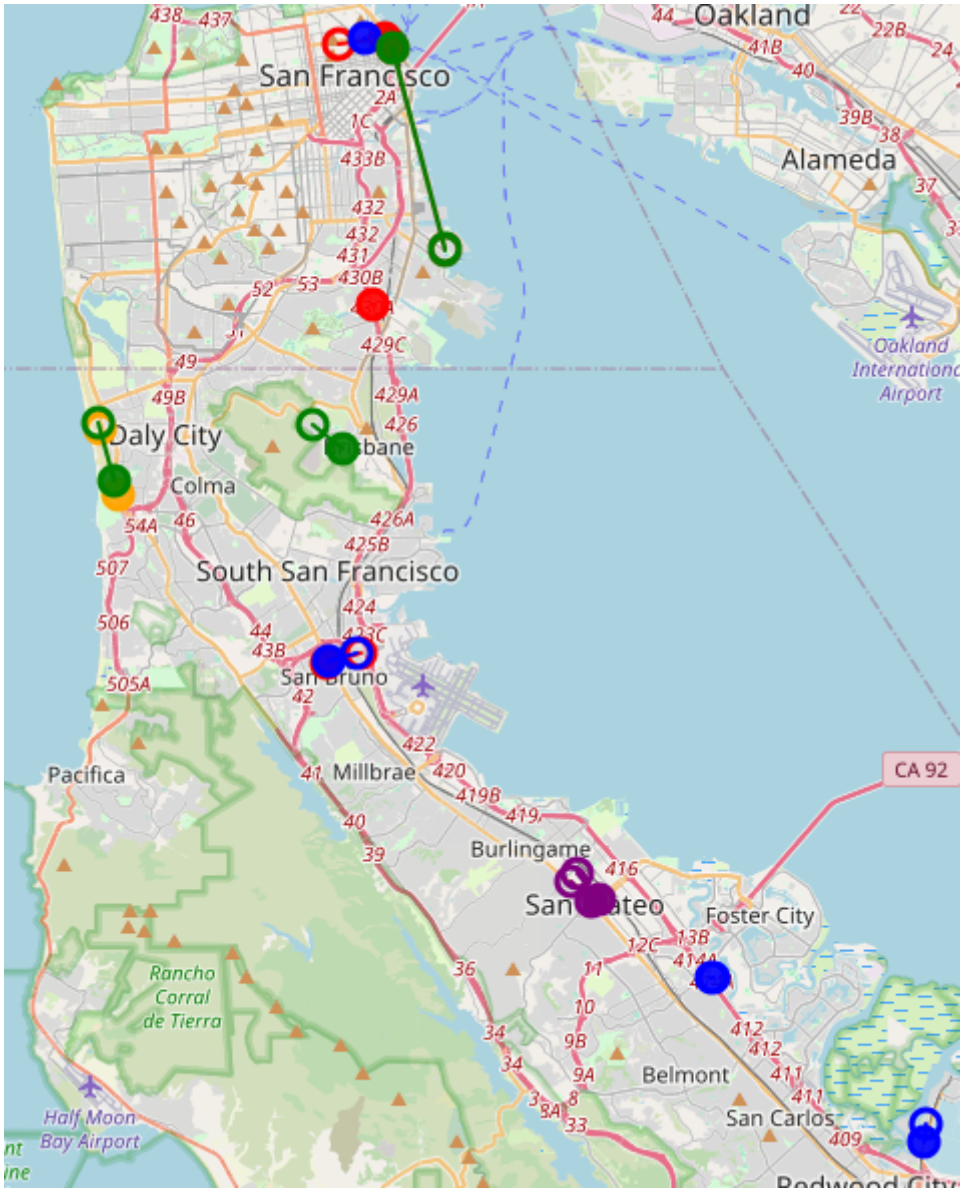

Figure 1: Tie record pairs across the Bay Area sample, with Haversine distances labeled

How should ties be resolved? In previous work, I resolved similar issues by cross-referencing multiple geocoders, ranking their outputs, and selecting either the most credible or the geographic midpoint.

4 Next Steps

- Use google maps API to fill remaining missing adresses

5 Appendix

Link to code repository:

- https://github.com/WilliamClintC/Geocoding_Office